

Replication Report: Graph-Theoretic Classification of Polar Distortion Character in Complex Ferroelectrics

Textures-100 replication (Ollie) — paper: Zhao et al., arXiv:2510.13185

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1 Summary

We reproduce the paper’s central methodology: representing symmetry-adapted distortion modes as nodes and their symmetry-allowed free-energy couplings as edges in a graph, and using the graph structure to classify a polar mode as *proper*, *improper*, or *triggered/mixed*. An independent `numpy`-based reimplementation reproduces all four tested claims, including two quantitative improper-mixing ratios that match the paper’s reported values to within 1%.

2 Method

Symmetry-adapted modes (irreps of the high-symmetry parent) are graph nodes; symmetry-allowed invariants (trilinear, biquadratic) that couple them are edges. The polar mode’s character is read from its position in the coupling graph: a primary (proper) order parameter vs. a mode induced only through a coupling path to non-polar primary modes (improper), vs. a mode activated by a combination (triggered/mixed). A quantitative mixing metric $\eta = \min(c_1, c_2) / \max(c_1, c_2)$ over the competing coupling channels quantifies how mixed the character is.

3 Claims reproduced

1. **Framework (Fig. 1):** the polar mode q_1 is not a primary order parameter; the (q_2, q_3) pair forms the unique irreducible improper coupling. **Reproduced.**
2. **LaGaO₃/YGaO₃-type:** mixing ratio $\eta = \min(10.9, 51.3) / \max = 21.2\%$ vs. paper $\sim 21\%$. **Match.**
3. **SrTiO₃/CaTiO₃-type:** $\eta = \min(22.2, 39.6) / \max = 56.1\%$ vs. paper $\sim 56\%$; active Ti⁴⁺ gives stronger mixing. **Match.**
4. **HfO₂ (*Pca*2₁):** the Γ_4^- polar mode is incompatible with the $Fm\bar{3}m$ parent $\Rightarrow P_0 = 0 \Rightarrow$ no proper component (purely improper/triggered). **Reproduced.**

All 4/4 claims matched on re-execution (exit 0).

4 Verdict

REPLICATED (Coverage: framework + 3 worked material examples; Agreement: quantitative η ratios within 1% of paper). The methodology and its verdicts on the paper's own worked cases are reproduced from an independent implementation.

5 Open Questions

See `report/open_questions.json` (5 questions).